

Shikun Sun

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My primary research focus is on diffusion models, with a specific interest in advancing new theoretical frameworks, enhancing controllable generation, exploring RL-based fine-tuning techniques, and improving computational efficiency. **My ultimate goal is to apply algorithms to create useful tools for artists.**

Education

- Tsinghua University** Bachelor Degree
Computer Science and Technology 2017.09 - 2021.06
– Undergraduate Thesis: *Aesthetic Knowledge Guided Image Sentiment Transfer*
- Tsinghua University** Second Bachelor Degree
Math and Applied Math 2018.09 - 2021.06
– Undergraduate Thesis: *Application of Invertible Neural Networks in Image Processing*
- Tsinghua University** Pursuing Doctoral Degree
Multimedia Information Processing 2021.09 - now
– Expected to graduate in 2026.06

Grade-Point Average

- Bachelor of Computer Science and Technology** 3.75
- Bachelor of Math and Applied Math** 3.26
- Ph.D. in Multimedia Information Processing (by now)** 3.89

Social Practice Activities

- Tsinghua University Bauhinia Volunteer Service Team** *Member*

Selected Award

- Tsinghua University Academic Excellence Award** 2018

Current Work

- Minimal Impact ControlNet (MIC):** This approach aims to improve the composition of multiple ControlNets by focusing on three key aspects: data augmentation, feature combination strategies inspired by multi-objective optimization, and ensuring score conservativity. These elements are designed to minimize negative interactions between ControlNets when used together, thereby achieving better overall control and stability in the network's output. *Internship at Alimama.*
- RL for Diffusion:** We applied DPO to Diffusion Models using the DPO with a novel CLIP method and a simple latent mix. This approach resulted in over a 10% improvement in real-world applications for the Wanxiang service, including various scenarios such as layout-to-image generation. *Internship at Alimama.*

Publications

My research publications since 2023 have been centered on the theory and application of Diffusion Models.

- Ziyi Wang, Xingqi Wang, Zeyu Jin, Xiaohan Li, **Shikun Sun**, Jia Jia. *AI Carpet: Automatic Generation of Aesthetic Carpet Pattern*. In Proceedings of the 30th ACM International Conference on Multimedia (MM'22)

- Zijie Ye, Jia Jia, Haozhe Wu, Shuo Huang, **Shikun Sun**, Junliang Xing. *Salient Co-Speech Gesture Synthesizing with Discrete Motion Representation*. In Proceedings of 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP'23)
- **Shikun Sun**, Jia Jia, Haozhe Wu, Zijie Ye, Junliang Xing. *MSNet: A Deep Architecture using Multi-Sentiment Semantics for Sentiment-Aware Image Style Transfer*. In Proceedings of the 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP'23)
- **Shikun Sun**, Longhui Wei, Junliang Xing, Jia Jia, Qi Tian. *SDDM: Score-Decomposed Diffusion Models on Manifolds for Unpaired Image-to-Image Translation*. In Proceedings of the Fortieth International Conference on Machine Learning. (ICML'23)
- **Shikun Sun**, Longhui Wei, Zhicai Wang, Zixuan Wang, Junliang Xing, Jia Jia, Qi Tian. *Inner Classifier-Free Guidance and Its Taylor Expansion for Diffusion Models*. In Proceedings of the Twelfth International Conference on Learning Representations. (ICLR'24)
- Zixuan Wang, Jia Jia, **Shikun Sun**, Haozhe Wu, Rong Han, Zhenyu Li, Di Tang, Zhou Jiaqing, Jiebo Luo. *DanceCamera3D: 3D Camera Movement Synthesis with Music and Dance*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024. (CVPR'24)
- Shuo Huang, **Shikun Sun**, Zixuan Wang, Xiaoyu Qin, Yanmin Xiong, Yuan Zhang, Pengfei Wan, Di Zhang, Jia Jia. *PlacidDreamer: Advancing Harmony in Text-to-3D Generation*. In Proceedings of the 32th ACM International Conference on Multimedia (MM'24)
- Zixuan Wang, Jiayi Li, Xiaoyu Qin, **Shikun Sun**, Songtao Zhou, Jia Jia, Jiebo Luo. *DanceCamAnimator: Keyframe-Based Controllable 3D Dance Camera Synthesis*. In Proceedings of the 32th ACM International Conference on Multimedia (MM'24)